Introduction

TAT-QA (**T**abular **A**nd **T**extual dataset for **Q**uestion **A**nswering) is a large-scale QA dataset, aiming to stimulate progress of QA research over more complex and realistic tabular and textual data, especially those requiring numerical reasoning.

The unique features of TAT-QA include:

- The context given is hybrid, comprising a semi-structured table and at least two relevant paragraphs that describe, analyze or complement the table;
- The questions are generated by the humans with rich financial knowledge, most are practical;
- The answer forms are diverse, including single span, multiple spans and free-form;
- To answer the questions, various numerical reasoning capabilities are usually required, including addition (+), subtraction (-), multiplication (x), division (/), counting, comparison, sorting, and their compositions;
- In addition to the ground-truth answers, the corresponding derivations and scale are also provided if any.

In total, TAT-QA contains **16,552** questions associated with **2,757** hybrid contexts from real-world financial reports.

The following is an example of TAT-QA. The left dashed line box shows a hybrid context. The rows with blue background are row header while the column with grey is column header. The right solid line box shows corresponding question, answer with its scale, and derivation to arrive at the answer.

Revenue from external customers, classified by significant product and service offerings, was as follows:

(in **millions**)			
Year Ended June 30,	2019	2018	2017
Server products and cloud services	32,622	26,129	21,649
Office products and cloud services	31,769	28,316	25,573
Windows	20,395	19,518	18,593
Gaming	11,386	10,353	9,051
Search advertising	7,628	7,012	6,219
LinkedIn	6,754	5,259	2,271
Enterprise Services	6,124	5,846	5,542
Devices	6,095	5,134	5,062
Other	3,070	2,793	2,611
Total	$125,843	$110,360	$96,571

Our commercial cloud revenue, which includes Office 365 Commercial, Azure, the commercial portion of LinkedIn, Dynamics 365, and other commercial cloud properties, was $38.1 billion, $26.6 billion and $16.2 billion in fiscal years 2019, 2018, and 2017, respectively. These amounts are primarily included in Office products and cloud services, Server products and cloud services, and LinkedIn in the table above.

#	Reasoning	Question	Answer	Scale	Derivation
1	Word Matching (38.06%)	How much revenue came from Linkedin in 2018?	5,259	million	-
2	Set of spans (11.94%)	Which were the bottom 2 revenue items for 2017?	LinkedIn, Other	-	-
3	Comparison (5.65%)	Which year has the lowest revenue?	2017	-	-
4	Counting (2.28%)	How many revenue items are between 6,000 million and 6,500 million in 2019?	2	-	Devices ## Enterprise Services
5	Addition (2.37%)	What is the total revenue of commercial cloud from 2017 to 2018?	42.8	billion	26.6 + 16.2
6	Subtraction (16.17%)	How much of the total revenue in 2018 did not come from devices?	105,226	million	110,360 - 5,134
7	Division (3.84%)	How much does the commercial cloud revenue account for the total revenue in 2019?	30.28	%	38.1 billion / 125,843 million
8	Composition (19.69%)	What was the percentage change in gaming between 2018 and 2019?	9.98	%	(11,386 - 10,353) / 10,353

For more information, please read our ACL 2021 paper [PDF] (https://aclanthology.org/2021.acl-long.254.pdf). TAT-DQA (https://nextplusplus.github.io/TAT-DQA/) is a new large-scale Document Visual QA (VQA) dataset, which is constructed by extending the TAT-QA. Please check out it if you are interested in the new task.

Getting Started

Download a copy of the dataset in json format:

- Training set (https://github.com/NExTplusplus/tat-qa/blob/master/dataset_raw/tatqa_dataset_train.json) [MD5: cc5026bdfe51bb47d63e6c3d31714951]
- Dev set (https://github.com/NExTplusplus/tat-qa/blob/master/dataset_raw/tatqa_dataset_dev.json) [MD5: 0b24a68b35fd814df5ad12cba548a8ea]
- Test set (https://github.com/NExTplusplus/tat-qa/blob/master/dataset_raw/tatqa_dataset_test.json) [MD5: c6ccc2beecaed12fc070c3df102ca019]

```
{
  "table": {                                                    # The tabular data in a
    "uid": "3ffd9053-a45d-491c-957a-1b2fa0af0570",              # The unique id of a ta
    "table": [                                                  # The table content whi
      [
        "",
        "2019",
        "2018",
        "2017"
      ],
      [
        "Fixed Price",
        "$  1,452.4",
        "$  1,146.2",
        "$  1,036.9"
      ],
      ...
    ]
  },
  "paragraphs": [                                               # The textual data in
    {
      "uid": "f4ac7069-10a2-47e9-995c-3903293b3d47",            # The unique id of a p
      "order": 1,                                               # The order of the par
      "text": "Sales by Contract Type: Substantially all of     # The content of the p
       our contracts are fixed-price type contracts.
       Sales included in Other contract types represent cost
       plus and time and material type contracts."
    },
    ...
  ],
  "questions": [                                                # The questions associ
    {
      "uid": "eb787966-fa02-401f-bfaf-ccabf3828b23",            # The unique id of a q
      "order": 2,                                               # The order of the que
      "question": "What is the change in Other in 2019 from 2018?",     # The question itself
      "answer": -12.6,                                          # The ground-truth ans
      "derivation": "44.1 - 56.7",                              # The derivation that
      "answer_type": "arithmetic",                              # The answer type incl
      "answer_from": "table-text",                              # The source of the an
      "rel_paragraphs": [                                       # The orders of the pa
        "2"
      ],
      "req_comparison": false,                                  # A flag indicating if
      "scale": "million"                                        # The scale of the ans
    }
  ]
}
```

Leaderboard

Rank	Model Name	Team Name	Exact Match	F1	Created	Paper
-	Human Performance	-	84.1	90.8	-	-
1	TAT-LLM (70B)	NExT	**81.4**	**88.4**	20 Jan 2024	Paper (https://arxiv.org/pdf/2401.
2	MATATA-8B	Docugami	77.6	84.9	9 Dec 2024	Paper (https://arxiv.org/pdf/2411.
3	TAT-LLM (13B)	NExT	77.5	85.9	20 Jan 2024	Paper (https://arxiv.org/pdf/2401.
4	Code Generation for Table-Text Question using LLM (70B)	Anonymous	76.8	84.7	21 Sep 2023	N.A.
5	TAT-LLM (7B)	NExT	76.4	85.1	20 Jan 2024	Paper (https://arxiv.org/pdf/2401.
6	AeNER: Attention-enhanced Numerical Embeddings for Reasoning	Gryffindor	75.0	83.2	16 May 2022	N.A.
7	MATATA-3.8B	Docugami	74.2	82.4	9 Dec 2024	Paper (https://arxiv.org/pdf/2411.
8	Code Generation for Table-Text Question using LLM (13B)	Anonymous	73.7	81.8	21 Sep 2023	N.A.
9	Encore	HIT-SCIR	71.8	80.1	24 Oct 2022	Paper (https://aclanthology.org/2
10	KFEX-N: A Table-Text QA Model with Knowledge-Fused Encoder & EX-N Tree Decoder	CWQian China	71.0	79.5	29 Oct 2023	Paper (https://www.sciencedirect.com/s

Rank	Model Name	Team Name	Exact Match	F1	Created	Paper
11	MVGE: Multi-View Graph Encoder for Answering Hybrid Numerical Reasoning Question	Weiyifan@CASIA	70.9	79.1	23 Dec 2022	Paper (https://arxiv.org/pdf/2305.
12	RegHNT: Relational graph neural network with special multitask decoder	LFyimi@CASIA China	70.3	77.9	5 May 2022	Paper (https://arxiv.org/pdf/2209.
13	Code Generation for Table-Text Question using LLM (7B)	Anonymous	68.4	77.3	21 Sep 2023	N.A.
14	UniRPG: Unified Discrete Reasoning over Table and Text as Program Generation	JD AI Research	67.2	76.0	24 Feb 2022	Paper (https://arxiv.org/pdf/2210.
15	RSTQA: Rounds Specified numerical reasoning for Table-Text QA	NLP2CT@UM Macau	66.8	75.0	30 Oct 2023	N.A.
16	SoarGraph: Semantic-Oriented Hierarchical Graphs	NExT	65.4	75.3	8 Sep 2022	Paper (https://dl.acm.org/doi/abs
17	UniPCQA	NExT / CUHK	63.9	72.2	22 Oct 2022	Paper (https://arxiv.org/pdf/2210.
18	MHST: Multi-Head with Sequence to Expression Tree	NExT	63.6	72.7	24 May 2022	Paper (https://dl.acm.org/doi/abs
19	GANO: GNN for Tabular and Textual QA with Numerical Reasoning	iLab@AIST Japan	61.9	72.1	15 Jul 2022	Paper (https://aclanthology.org/2

Rank	Model Name	Team Name	Exact Match	F1	Created	Paper
20	TBC	Anonymous	60.8	68.7	22 Jun 2022	N.A.
21	FinMath: Injecting a Tree-structured Solver for Question Answering over Financial Reports	FinMath@NEU China	58.3	68.2	6 Aug 2022	Paper (http://www.lrec-conf.org/p 1.661.pdf)
22	KIQA: Knowledge-infused QA Model for Table and Text	iLab@AIST Japan	58.2	67.4	23 Feb 2022	Paper (https://aclanthology.org/2
23	LETTER: Logic Enhanced Table-Text Reasoning	OnceAgain	56.1	64.3	17 Feb 2022	N.A.
24	TeaBReaC-pretrained T5-3B	SBU / Allen AI	55.8	63.8	17 Jun 2022	Paper (https://arxiv.org/pdf/2205.
25	OPERA-H	Hero_Dirk	55.2	63.8	9 Oct 2022	N.A.
26	GenQA:Generative model for QA from table and text	IITJ@India	55.1	65.6	7 Feb 2023	N.A.
27	Baseline - TagOp	NExT	50.1	58.0	13 May 2021	Paper (https://aclanthology.org/2

Submission

To evaluate your models, we have also made available the evaluation script we will use for official evaluation, To run the evaluation, use

```
python tatqa_eval.py --gold_path=:path_to_dev --pred_path=:path_to_predictions
```

Predictions Format

The predictions file in JSON format contains a dictionary with question ids as keys and the predictions as values (each prediction shall include both `answer` and `scale` in an array). For example,

```
{
 "9337c3e6-c53f-45a9-836a-02c474ceac16": [
    "4.6",
    "percent"
  ],
  "c4170232-e89c-487a-97c5-afad45e9d702": [
    "16",
    "thousand"
  ],
  "d81d1ae7-363c-4b47-8eea-1906fef33856": [
    ["2018", "2019"],
    ""
  ]
  ...
}
```

We also provide a sample prediction file (https://github.com/NExTplusplus/tat-qa/blob/master/sample_prediction.json) (on Dev) for your reference.

```
python tatqa_eval.py --gold_path=dataset_raw/tatqa_dataset_dev.json --pred_path=sample_predicti
```

Submission

Please email the prediction file of the test set with the following information to us:

- Model name with a brief description of your model
- Team name
- Contact person
- Email of the contact person
- Link to the paper (if any)
- Link to the code repository (if any)

Please give us up to two weeks to evaluate your submission and we will add your model to the leaderboard.

Contact

The TAT-QA dataset is under the license of Creative Commons (CC BY) Attribution 4.0 International. (https://creativecommons.org/licenses/by/4.0/)

For more information, please contact the author: Fengbin ZHU fengbinzhu@u.nus.edu (mailto:fengbinzhu@u.nus.edu)

Please kindly cite our work if you use our dataset or codes, thank you.

```
@inproceedings{zhu2021tat,
    title = "{TAT}-{QA}: A Question Answering Benchmark on a Hybrid of Tabular and Textual Cont
    author = "Zhu, Fengbin  and
      Lei, Wenqiang  and
      Huang, Youcheng  and
      Wang, Chao  and
      Zhang, Shuo  and
      Lv, Jiancheng  and
      Feng, Fuli  and
      Chua, Tat-Seng",
    booktitle = "Proceedings of the 59th Annual Meeting of the Association for Computational Li
    month = aug,
    year = "2021",
    address = "Online",
    publisher = "Association for Computational Linguistics",
    url = "https://aclanthology.org/2021.acl-long.254",
    doi = "10.18653/v1/2021.acl-long.254",
    pages = "3277--3287"
}
```

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